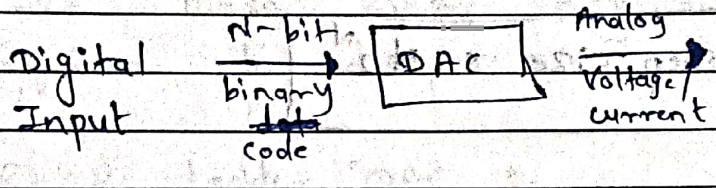
✱ Chapter - 05 ✱

✱ DACs and ADCs ✱

✱ DAC [Digital to Analog converter] :-

- A DAC converts a digital binary input into a corresponding analog voltage or current.

• Block Diagram :-



eg. Suppose we have 3-bit DAC

Digital input	→	Analog output
000	→	minimum
001	→	slightly higher
010	→	higher
011	→	higher
100	→	higher
101	→	higher
110	→	higher
111	→	maximum

∴ A 3-bit DAC has :-
$2^3 = 8$ possible input codes

• DAC resolution :-

- Resolution tells us how many different digital levels the DAC can represent.

- For N-bit DAC → Number of levels $= 2^N$

- Higher resolution → smaller steps → more precise representation

• ⊛ Reference Voltage (V_{REF}) :-

- The DAC uses a reference voltage to determine its output range.

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eg. 8-bit DAC → $V_{ref} = 3.3\ V$

- The ~~input~~ digital input code determines what fraction of the reference voltage appears at the output.

$$V_{ref} = \text{step size} \times 2^N$$

- LSB / step size :-
  - for n-bit DAC :-

$$LSB \approx \frac{V_{ref}}{2^n}$$

- Analog output :-
  - The digital input determines the analog output
  - A simplified ideal DAC relationship is :-

$$V_{out} \approx \frac{\text{code}}{2^N} \times V_{ref}$$

- DAC Architectures :-

① Binary-weighted Resistor DAC :-
- Uses resistors weighted accordingly to binary bit

eg.

MSB → larger contribution
- → bit 2
- → bit 1
- → LSB → smaller contribution

- Each bit contributes a different amount to the analog output

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R-2R DAC → only R and 2R values

- Note :-
  - In case of designing of weighted register type DAC, we require perfectly matched registers which practically very difficult as number of resistance goes on increasing. That's why we design R-2R ladder where only two resistors are required to design DAC of any no. of bits.

② R-2R ladder DAC :-
- It uses only two resistor values :- i.e. R, 2R
- This resistor network form a ladder
- only two register values are required; making the architecture easier to scale and match than a binary-weighted resistor network

③ Current-steering DAC
- Instead of primarily using resistor networks, it uses current sources / switches to generate the output.
- It is commonly associated with high-speed DAC applications

- Some basic Terms :-

① Accuracy :-
- How close the actual analog output is to the expected / ideal output

② DNL - Differential Non-Linearity
- DNL describes how much the actual step between adjacent codes differs from the ideal 1-LSB step

③ INL — Integral Non-Linearity

- INL describes how far the actual transfer characteristics deviates from the ideal transfer characteristics

④ Offset Error

- Offset error means the DAC has an error in its output even around the starting / zero-code region.
- Ideally → Code = 0 → Vout ≈ 0V
- If there is → Code = 0 → Vout ≠ ideal 0V offset error

⑤ Gain Error :-

- Gain error is the difference in the slope / full-scale behaviour between the actual DAC transfer characteristics and the ideal one, after accounting for offset
- Simple Idea :→

  Ideal : 0 → 3.3 V

  Actual : 0 → 3.2 V

⑥ Settling Time :-

- Settling Time is the time required for DAC output to reach and remain within a specified error band around its final value after an input code change.

⑦ Monotonicity :-

- A DAC is monotonic if increasing the digital input code does not cause the analog output to decrease

⑧ DAC Digital Interface :-

- A typical DAC often has a digital interface through which software / RTL programs the DAC.
- Common Interfaces :-
  1. SPI
  2. I^2C
  3. Parallel Interface

⑨ DAC Registers :-

- A DAC device may contain registers such as :
  1. Control Register
  2. Data Register
  3. Status Register
  4. Configuration Register

* ADC [Analog to Digital converter] :-
- An ADC converts an analog input voltage into a digital binary value.

- The exact output depends on ADC's resolution, reference voltage and coding scheme.

• Resolution :-
- Resolution is no. of bits used by an ADC to represent the input.
  eg. for n-bit ADC → No. of codes = 2^n

• Vref and LSB :-
- Vref defines the reference / input range used by the ADC
  eg. Vref = 5V → I/p range ≈ 0V to 5V

- LSB represents the nominal voltage step between ADC codes.
  for an ideal N-bit ADC :
  $$LSB = \frac{V_{ref}}{2^n}$$

• ADC transfer function :-
- The transfer function describes the relationship bet^n Input voltage and ADC output code
- Input voltage ↑ → o/p code ↑

• Sampling :-
- Sampling means ~~talking~~ taking measurement of an analog signal at specific time intervals

- Sampling freqⁿ :-
  - No of samples taken per seconds

$$f_s = \frac{1}{T_s}$$

- Nyquist Theorem :-
  - for a signal with max freqⁿ

$$f_s > 2 f_{max}$$

- Quantization
  - Quantization converts a sampled analog voltage into one of a finite number of digital levels
  - Because the ADC has finite resolution, the converted value cannot exactly represent every possible analog voltage.

- Quantization Error :-
  - Difference betⁿ the actual sampled value and the ideal quantized value.

$$\therefore \text{Error} = \pm \frac{1}{2} \text{ LSB}$$

- ADC Architecture :-
  1. SAR ADC
  2. Flash ADC
  3. Pipline ADC
  4. Sigma-Delta ADC

1. SAR ADC [Successive Approximation Register]
   - It determines the output bits one by one, generally from MSB to LSB

Input → Comparator → SAR logic → Digital O/p

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② Flash ADC
- Uses many comparators working in parallel

③ Pipeline ADC
- Conversion is divided into multiple stages

④ Sigma-Delta ADC
- Uses oversampling, noise shaping & digital filt

• ADC specifications :-

① DNL [Differential Non-Linearity]
- shows how much an individual code width differs from the ideal code width

② INL [Integral Non-Linearity]
- shows how much the overall transfer curve deviates from the ideal

③ offset Error
- Difference bet^n the actual and ideal transfer characteristics near the specified midscale transi

④ Gain Error
- Diff. in the slope of actual transfer charac eristics from the ideal one after accounting for offset, according to the specified definition

⑤ SNR [Signal to Noise Ratio]

$$SNR = 10 \log_{10} \frac{P_{signal}}{P_{noise}}$$